

Generating language and decoding brains

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ABSTRACT. Can we read other people’s minds by seeing the neuroimaging data of their brains? Recent discussion claims success, but success is self-reinforced by design. More data and better statistics yield a better approximation to a mass of unanalyzed data. But approximation of language production is essentially different from explanation of language generation. Causal explanations of generative grammar may be confirmed by some statistical experiment; in contrast, a statistical result must still be causally explained before some theoretical understanding can be reached. Probability theory is incapable in principle of advancing scientific understanding beyond a measure of prediction based on statistical calculations that have no neurobiological reality in human brains.

Key words neuroimaging, brain decoding, reverse inference, semantic vectors, prediction vs. explanation, statistical approximation vs. Merge computation, possible vs impossible languages

1. Introduction. As early as 2008 *Science* published an article proposing a computational model for inferring brain activity from content words. The model related the statistics of word co-occurrence in text to fMRI neural activation associated with thinking about word meanings (Mitchell et al., 2008).¹ A decade later, Fedorenko and her team presented a model that works the other way round (Pereira et al., 2018).² The goal of their experiment is the decoding of mental states from neuroimaging data (“reverse inference”). A brain decoding system is trained on imaging data and learns which brain patterns are evoked by which words, and how to use that knowledge to infer which words are implied by novel brain activation patterns it encounters, thus inferring meaning from neuroimaging. In general, decoding methods opportunistically aim at predictive power instead of neurobiological reality, following a kind of “abductive inference” or “reasoning to the best explanation” (Poldrack, 2011).³ Furthermore, neuroimaging data are “inherently correlational” and, hence, far removed from a causal explanation of how the brain synthesizes mental states from neuroimaging inputs.

2. Decoding the brain. The Pereira experiment is based on the idea that “[p]atterns of activation for semantically related stimuli are more similar to each other than for unrelated stimuli.” It would then follow that “[v]ariation in each semantic space dimension would correspond to variation in patterns of activation.” (Pereira et al., 2018).⁴ The brain decoding device exploits the correspondence to learn the relationship between the two. The decoder is trained on brain activation patterns (fMRI images) and corresponding text-derived semantic vectors and is tested on new brain imaging data to decode semantic vectors. The device, a computer program, calculates variations in the new activation pattern relative to patterns from learned experience and infers a quantitative representation of mental content from it --- a semantic vector. The decoded vectors are then compared with their text semantic vectors, where a text-derived vector for a sentence is constructed by averaging its content word vectors. Success is claimed when the semantic vector decoded from a brain activation pattern best approximates its text-derived semantic vector. That’s the case if the highest correlation is between the decoded vector and the corresponding text-derived semantic vector.

¹ Mitchell, T.M., S.V. Shinkareva, A. Carlson, K.M. Chang, V.L. Malave, R.A. Mason, M.A. Just. 2008. Predicting human brain activity associated with the meanings of nouns. *Science* 320(5880):1191-1195. doi: 10.1126/science.1152876

² Pereira, F, B. Lou, B. Pritchett, S. Ritter, S.J. Gershman, M. Botvinick, and E. Fedorenko. 2018. Toward a universal decoder of linguistic meaning from brain activation. *Nat Commun* 9, 963. <https://doi.org/10.1038/s41467-018-03068-4>

³ Poldrack, R.A. 2011. Inferring mental states from neuroimaging data: From reverse inference to large-scale decoding. *Neuron* 72(5), 692-697.

⁴ Pereira, F. et al (2018) op. cit.

3. Success and failure by design. How does the experiment work in slightly more detail? We introduce some terminology. Let $R(P_x)$ be the imaging data (fMRI) that represents brain activation pattern P_x . $R(C_x)$ then represents mental content C_x activated in P_x and is the text-derived semantic vector of $R(P_x)$. The pairs $\langle R(P_x), R(C_x) \rangle$ thus (partially) represent phonetic form (sensorimotor) and semantic interpretation (conceptual-intentional), i.e., the pairs $\langle CI, SM \rangle$ of generative models of language (Chomsky, 2021, 2024).⁵ Still, there's an important difference, approximation by statistics (describing) vs. proof by recursive computation (explaining).

A “universal” decoder UD computes a semantic vector from $R(P_x)$. I.e., $UD(R(P_x)) = R(C_x')$. The meaning of a word is represented as a vector in a 300-dimensional semantic space constructed from massive text corpora; the meaning of a phrase or sentence is represented as an *average* of the meanings of its content words. The decoded semantic vector $R(C_x')$ is then pairwise tested against other (decoded, text-derived) semantic vectors. The test is considered successful if the highest correlation is between $R(C_x)$ and $R(C_x')$, i.e., $R(C_x) \cong R(C_x')$.

The question then arises: Will $R(C_x')$, decoded from $R(P_x)$ of *Dog bites man*, approximate $R(C_y')$, decoded from $R(P_y)$ for *Man bites dog*? Will decoded vectors $R(C_x')$ and $R(C_y')$ capture a “similarity of structure” in their semantic representations? The sentences contain the same content words but have a different theta structure, “invisible” to the decoder that is not trained on syntactic phrase structure. If both decoded vectors $R(C_x')$ and $R(C_y')$ “approximate” the text-derived vector $R(C_x)$ for $R(P_x)$ reasonably well, not only has no causal explanation been offered --- since a correlational description is the best possible result ---, but also the descriptive attempt fails, yielding an incorrect result by design (averaging semantic vectors of words for propositional meaning without any information from phrase structure).

4. Prediction in need of explanation. Using powerful statistical techniques, semantic vectors are decoded from imaging data that best approximate corresponding text semantic vectors. These tools aim at highest matching probability (much like Google neural machine translation, or deep learning GPT3), steering away from explanation of what is going on in language processing. However, true explanation goes beyond mere description and prediction and answers the question of why things work the way they do and not otherwise. Science must yield interpretable explanations, not unexplained predictions (Chomsky, 2020, 2021, 2022, 2024; Kodner et al., 2023; Vázquez Martínez et al., 2024; Chomsky et al., 2023a).⁶ Scientific understanding, therefore, requires a neurobiological model of the brain, not a probabilistic approximation of data stored in memory. Perhaps an important technological development with useful prospective applications, the “universal” brain decoder yields only a correlational, not a causal, account and, therefore, cannot be a realistic model of what goes on in the human brain. But a more hopeful perspective could perhaps be realized eventually. For valuable discussion of how today's prevailing mechanistic and associationist deep learning programs could develop into tomorrow's true cognitive science that more adequately explains human understanding we refer the reader to Bever (2021).⁷

On the other hand, there already exists an explanatory computational account: a universal theory of I(nternal)-language that deterministically links meaning to sound and determines (non-)identity and degrees

⁵ Chomsky, N. 2021. Minimalism: Where are we now, and where can we hope to go. *Gengo Kenkyo* 160: 1-41; Chomsky, N. 2024. Genuine explanation. In: Giuliano Bocci, Daniele Botteri, Claudia Manetti, and Vincenzo Moscati (eds.), *Rich Descriptions and Simple Explanations in Morphosyntax and Language Acquisition*. Oxford, UK: Oxford University Press, 15-44.

⁶ Chomsky, N. (2021) op. cit.; Chomsky, N. (2024) op. cit.; Chomsky, N. 2020. Simplicity and the Form of Grammars. *Journal of Language Modelling* 9(1): 5-15; Chomsky, N. 2022. Genuine explanation and the strong minimalist thesis. thesis. *Cognitive Semantics* 8(3): 347-365; Chomsky, N., I. Roberts, and J. Watumull. 2023a. “Noam Chomsky: The False Promise of ChatGPT.” *The New York Times*, 2023, sec. Opinion. <https://www.nytimes.com/2023/03/08/opinion/noam-chomsky-chatgpt-ai.html>; Kodner, J., S. Payne, J. Heinz. Why Linguistics Will Thrive in the 21st Century: A Reply to Piantadosi (2023), September 2023, [lingbuzz/007485](https://lingbuzz.net/lingbuzz/007485). https://ling.auf.net/lingbuzz/007485/current.pdf?_s=CXxeoHpFk_XRY0DG; Vázquez Martínez, H.J., A. Heuser, C. Yang, J. Kodner. Evaluating the Existence Proof: LLMs as Cognitive Models of Language Acquisition, July 2024, [lingbuzz/008277](https://lingbuzz.net/lingbuzz/008277). https://lingbuzz.net/lingbuzz/008277/current.pdf?_s=QJuWPMc4eVI4vXpL

⁷ Bever, T.G. 2021. How *Cognition* came into being. *Cognition* 213(2-3): 104761. DOI:10.1016/j.cognition.2021.104761

of relatedness between pairs of sensorimotor and semantic representations. Minimalist grammar based on the *Strong Minimalist Thesis* (SMT) provides us with a computational procedure explaining how meaning is determined from structure and externalization is realized when activated (Chomsky, 2020, 2021, 2022, 2024; Chomsky et al., 2023b).⁸ Neither language learning infants nor adults with internal language already fully in place use powerful statistical/associationist mechanisms for processing immense corpora of chaotic, unanalyzed data stored in memory only to approximate mental content. Instead, both are guided by an inborn basic program for UG-based computations constrained by organic third factor principles.

From this perspective at least, the experiment reflects an unrealistic model of the human brain. That's not necessarily a problem. It's a matter of choosing perspective. Do you want an explanatory theory of the mind/brain or a useful technological application with a fair measure of success? Two legitimate interests. Biologically plausible SMT-based parsers and deep learning neural networks for processing language are different but not competing programs. Problems only arise when the different goals are confused, and the programs are pitted against each other. Predictive success in decoding meaning from neuroimaging cannot *per se* be claimed as an argument against (merge-generated) syntactic structure of language. Conversely, explanatory results of (merge-based) syntax in generative grammar cannot *per se* be used to reject Google Translate. These are different games with different objectives and different rules of engagement.

5. Interpretable explanation vs unexplained approximation.

5.1. Conceptual considerations. Since both text-derived and decoded meaning vectors are computed from content words by design only, the model aims at producing the best possible match between brain activation patterns and semantic vectors. Interest is in degrees of acceptability rather than degrees of grammaticality. The model does not aim at explaining the internal mechanisms of language that sharply distinguish possible (*The reviews of his book annoy him* or *The review of his books annoys him*) from impossible languages (**The reviews of his book annoys him* or **The review of his books annoy him*). The distinction doesn't simply illustrate an idiosyncrasy of English grammar but rather exemplifies a universal property of all constructions in all languages, namely structure-dependence (Moro, 2008/2015, 2016; Chomsky, 2021, 2022, 2024).⁹ Human language proves sensitive to hierarchical structure and ignores linear order. A remarkable discovery of generative grammar that receives a natural explanation from set-theoretic binary Merge, Simplest Merge, the basic recursive operation of language that takes two syntactic objects X,Y and constructs from them a new set {X,Y}, without modifying either element or adding order or any other linear arrangement (see the references in footnote 8 for detailed discussion).

However, if syntax is ignored in decoding meaning from neuroimaging, there are yet other challenges to overcome. For example, the meanings of *Flying planes are dangerous* and *Flying planes is dangerous* are quite different and follow from the different syntactic structures that characterize these expressions. But since the meaning vectors of the sentences are averaged from the vectors of their content words only, the model will fail in principle to correlate different decoded vectors with different text-derived vectors. They will all be the same. The ambiguity of *Flying planes can be dangerous* will be lost, and with it an explanation for a basic property of human language, the structure-dependence of operations in a syntactic derivation (Chomsky, 1955/975, 1957).¹⁰ Externalization yields ambiguity for simple computational reasons: the internal language system allows the rules to apply freely without any concern for communicative efficiency.

⁸ Chomsky, N. (2020) op. cit.; Chomsky, N. (2021) op. cit.; Chomsky, N. (2022) op. cit.; Chomsky, N. (2024) op. cit.; Chomsky, N., T.D. Seely, R. Berwick, S. Fong, M.A.C. Huybregts, H. Kitahara, A. McInnerney, Y. Sugimoto. 2023b. *Merge and the Strong Minimalist Thesis*. In: R. Freidin (ed.), *Elements in Generative Syntax*. Cambridge, UK: Cambridge University Press.

⁹ Chomsky, N. (2021) op. cit.; Chomsky, N. (2022) op. cit.; Chomsky, N. (2024) op. cit.; Moro, A. 2008/2015. *The Boundaries of Babel: The Brain and the Enigma of Impossible Languages*. Cambridge, MA: MIT Press; Moro, A. 2016. *Impossible Languages*. Cambridge, MA: MIT Press.

¹⁰ Chomsky, N. 1955. *The Logical Structure of Linguistic Theory*. Ms. Harvard University, Cambridge, MA. and MIT, Cambridge, MA. [Revised 1956 version published in part by Plenum, New York, 1975, and University of Chicago Press, Chicago, 1985]; Chomsky, N. 1957. *Syntactic Structures*. The Hague: Mouton.

5.2. Empirical considerations. As a final illustration, the inherent and distributional properties of lateral fricative “l” in *verloopt* (“expires”) and guttural plosive “k” in *verkoopt* (“sells”) of Dutch sentences (1)-(2) are correlated with different syntactic and semantic structures but, clearly, cannot explain their relevant syntactic and semantic properties. At best, probabilistic approaches that calculate statistical regularities in word patterns may predict that (1b) or (2a) is less likely (i.e., more impossible/less possible) than (2b) or (1a).

- (1) a. de man z’n paspoort verloopt “the man’s passport expires”
 b. *de man verloopt z’n paspoort
- (2) a. *de man z’n paspoort verkoopt
 b. de man verkoopt z’n paspoort “the man sells his passport”

However, a better approximation of the data, i.e., an increasingly accurate prediction of the most likely possible word string next, does not offer a genuine explanation. Prediction based on statistical regularities differs fundamentally from explanation based on hypothetical-deductive argument. Higher probability of occurrence (yielding better approximation of acceptability) and potential explanation of an internal device (fully determining grammaticality) are two different and incommensurable categories.

The finite verb is traditionally said to be in second position (Verb-Second) of main clauses in continental West-Germanic. But, crucially, Verb-Second targets second *structural* position, not second *linear* position. In fact, the verbs (1a) and (2b) hold fifth resp. third digital position. The operation thus exclusively relies on structure. Structure-dependence is a hallmark of human language, which ignores the linear properties of word structure that constitute the basis of statistical calculation of LLMs that use deep learning to complete unfinished sentences or even generate complete sentences or texts. In minimalist generative grammar operations on word strings are illegitimate (“impossible”) because language is sensitive to hierarchical structure only (Everaert et al., 2015; Moro, 2016; Chomsky, 2021).¹¹

Verb-Second interacting with null topics further deepens the problem. The empirical data in (3) illustrate that the verbs in clause-initial position occupy second structural position. The ambiguous sentence *trok ze aan* (meaning “[that] she put on” or “[that] she pulled at”) has different stress patterns that only depend on a difference in hierarchical structure for phrasal verb phrase VP, namely [_{VP} NP [_V P V]] for (*ze*) *iets aan trok* [(“she) put something on”] vs. [_{VP} [_{PP} P NP] V] for (*ze*) *aan iets trok* [(“she) pulled at something”]. The first structural position of the sentences in (3) is filled by an empty element, the null topic, which binds a variable in argument position. The null topics of (3) are articulated (*dat*, *daar*) in the corresponding cases of (4); the bound variables are the direct object of “(iets) aan trok” in (4a) and prepositional object of “(ergens) aan trok” in (4b), as illustrated in the parenthesized clauses.

- (3) a. trok ze --- **áán** (cf. *ze trok --- áán) she put that on
 b. **trók** ze --- aan (cf. *ze trók --- aan) she pulled at that
- (4) a. *dat* trok ze -- **áán** (cf. ze trok *iets* áán) that she put on
 b. *daar* **trók** ze -- aan (cf. ze trók aan *iets*) that she pulled at

Expressions (3a, b) contain the same content words and have the same linear structure. Therefore, since their decoded meaning vectors would presumably be identical even if their text-derived vectors would not, the model would fail as a model of human language. Failure is due to a principled reason: the model is designed to answer the wrong question, ignoring the merge-based compositional syntax that would immediately have overcome the semantic problem.

¹¹ Moro, A. (2016) op. cit.; Chomsky, N. (2021) op. cit.; Everaert, M.B.H., M.A.C. Huybregts, N. Chomsky, B.C. Berwick, and J.J. Bolhuis. 2015. Structures, not strings: Linguistics as part of the cognitive sciences. *Trends Cogn. Sci.* 19, 729-743.

6. Further discussion. If, miraculously, decoded and text-derived vectors would perfectly match, the model can claim success, but the self-proclaimed success would not explain language because the model seeks to address the wrong question by design. The correct question that needs an answer is why language is the way it is and cannot be otherwise. What's the structure of (3a) and (3b)? How are they interpreted? And why are (1b)-(2a) or the parenthesized sentences in (3) ungrammatical? The infant's initial state of the faculty of language already entails the distinction between possible and impossible languages. Why would complex structure dependence and hierarchical complexity prevail if statistical models of simple linear structure could retrieve meaning? While LLMs can boast some success predicting a correlation, they fail to explain why this holds (Bever et al., 2023; Chomsky, 2023; Kodner et al., 2023; Vázquez Martínez et al., 2024; Chomsky et al., 2023a).¹² Suppose now that the empirical failures and experimental flaws in dealing with minimal pairs (1) vs (2) or (3a) vs (3b) would be overcome. What we would end up with is prediction of gradient acceptability, not an interpretable explanation of categorical grammaticality. Differences of structure and interpretation are left undiscussed. Approaching the limit between possible and impossible languages in terms of statistical regularities is quite different from genuine explanation as discussed in the references in footnotes 6 and 8. Approximation fails to explain properties of internal language because of a design flaw. Even if the model correctly predicts every possible/grammatical sentence to be more acceptable than any impossible/ungrammatical sentence for every human language, then there is still no explanation of their grammatical properties or how and why structure and interpretation correspond with the stress patterns of their linear structure. In contrast, answers immediately fall out from merge-based derivation constrained by SMT-principles.

7. Conclusion. Rather than computing meaning that best matches neuroimaging data, SMT'-constrained and UG-based derivation genuinely explains <CI, SM> pairs of I-language. No weak approximation but strong determination. Merge, the basic structure-building operation in human language generates an unbounded array of hierarchically structured expressions, each expression receiving its own CI representation and SM externalization (if activated). The two programs do not compete but have different aims. One is a generative procedure for constructing unbounded language that explains how meaning (CI) relates to phonetic form (SM) for each structure generated. The other program, probability theory, computes which semantic vector best matches which brain activation pattern. Best match is determined by strong algorithms that operate on statistical and associationist principles (frequency, contiguity, similarity, cause/effect) and which are grounded in statistical analysis of huge corpora of text and imaging data. The UG-based causal linking of phonetics and semantics of I-language is incomparably different from the weaker correlational linking which approximates linguistic meaning from decoding brain activation patterns. Causal explanations may be confirmed by empirical test, e.g., by statistical experiment; in contrast, a statistical result must still be causally explained before some theoretical understanding can be achieved. Probability theory is incapable in principle to advance scientific understanding beyond prediction based on statistical calculations that have no neurobiological reality in human brains. Although success is self-reinforcing by design since more data and better statistics will yield a better approximation to a mass of chaotic and unanalyzed data, a fair approximation to an immense corpus of text still doesn't explain anything about the underlying mechanism for generating human language.

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¹² Chomsky, N., I. Roberts, and J. Watumull (2023a) op. cit.; Kodner, J., et al. (2023) op. cit.; Vázquez Martínez et al. (2024) op. cit.; Bever, T.G., N. Chomsky, S. Fong, and M. Piattelli-Palmarini. 2023. Even deeper problems with neural network models of language, pp 24-25. doi:10.1017/S0140525X23001619, e387 [Commentary on Bowers, J.S. et al. 2023. Deep problems with neural network models of human vision. *Behavioral and Brain Sciences* 46, e385: 1–77. doi:10.1017/S0140525X22002813]; Chomsky, N. 2023. "ChatGPT and Human Intelligence: Noam Chomsky Responds to Critics." Noam Chomsky Interviewed by Ramin Mirfakhraie. April 24, 2023. *Monthly Review Online*. <https://mronline.org/2023/04/24/chatgpt-and-human-intelligence>